(natural language processing) and convolutional networks. Their multi-core efficiency makes them the foremost choice for deep end matrix calculations and can handle datasets even ranging to values of Terabytes.

And then comes the powerful Intel® Xeon™ Scalable system created on the 14nm process technology. They can support up to 28 physical cores (56 threads) per socket if using the Platinum series (up to 8 sockets). They even support six memory channels that can hold 1.5 TB space with DDR4 memory. The Intel® Xeon™ Scalable processor has been one of the popular choices for analysts. Projects for tracking price movements and predicting product classes in a production lot are just some places where the both CPUs and GPUs are used.

Intel introduced the AVX-512 instruction set that has been implemented on the latest Skylake-based CPUs. These can also be found on the Core-X series as well as the Xeon Scalable CPUs which also utilise the Skylake architecture. The industry has touted these instructions for some reasons for multi-scale regression analysis.

The AVX-512 instruction set can accelerate performance for heavy computational workloads that use a 512-bit wide Fused Multiply-Add (FMA) operations. This allows the system to conduct lower-precision operations with greater accuracy while also eliminating the need to retrain and resample datasets frequently.

To top it off, Intel also provides a Deep Learning Inference Engine as part of the Deep Learning Deployment Toolkit. The Inference Engine's current setup supports inference on Intel® Xeon™ with AVX2 and AVX512, Intel® Core ™ Processors with AVX2, Intel® Atom® Processors with SSE. And it also works well with Intel® HD Graphics. The Engine is a great tool for executing deep learning strategies in the FP16 and FP32 format.

LANGUAGES AND FRAMEWORKS

Data scientists have a great bunch of tools to their disposal that is used primarily for pushing the limits of the system in conducting more and more complex processes with less time and greater accuracy.

Intel has some tools and software that work well with accelerators. Software libraries such as nGraph, Intel® Math Kernel Library, Intel® Math Kernel Library for Deep Neural Networks, Intel® Machine Learning Scaling Library, Intel® Data Analytics Acceleration Library, Intel® Distribution for Python, etc., are some of the well optimised libraries out there right now. the hardware like the nGraph.